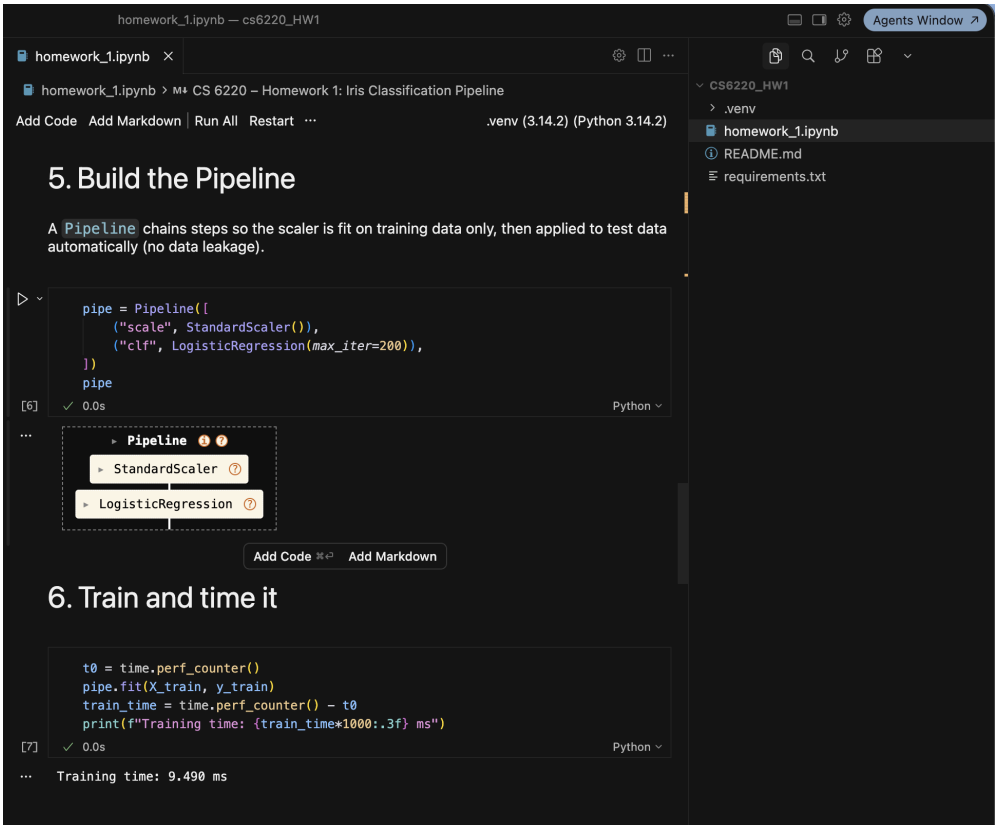
# CS 6220 Homework 1

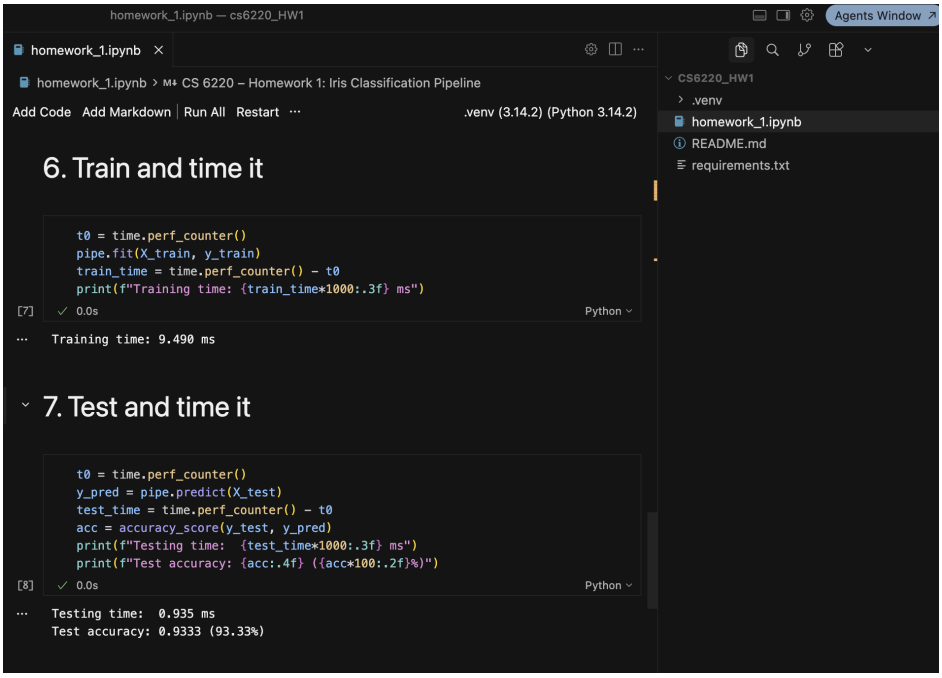
Jonathan Schwenk

1) None
2) https://github.com/JonathanSchwenk/cs6220_HW1
3)
    a) What programming language are you using?
        i) Python
    b) What tool do you use to write and debug the source code?
        i) I used Cursor for my IDE with Jupyter notebooks.
    c) What tool do you use to test and execute the code?
        i) Jupyter notebook kernel, ran inside Cursor.
    d) Are you using scikit-learn? If not, what are you using?
        i) Yes, I'm using scikit-learn, Pipeline, Logisticregression, StandardScaling, plus pandas and matplotlib.
    e) Are you running your program on your laptop? If not, which service are you using?
        i) Yes, my personal computer (Macbook)
4) Dev environment
    a)

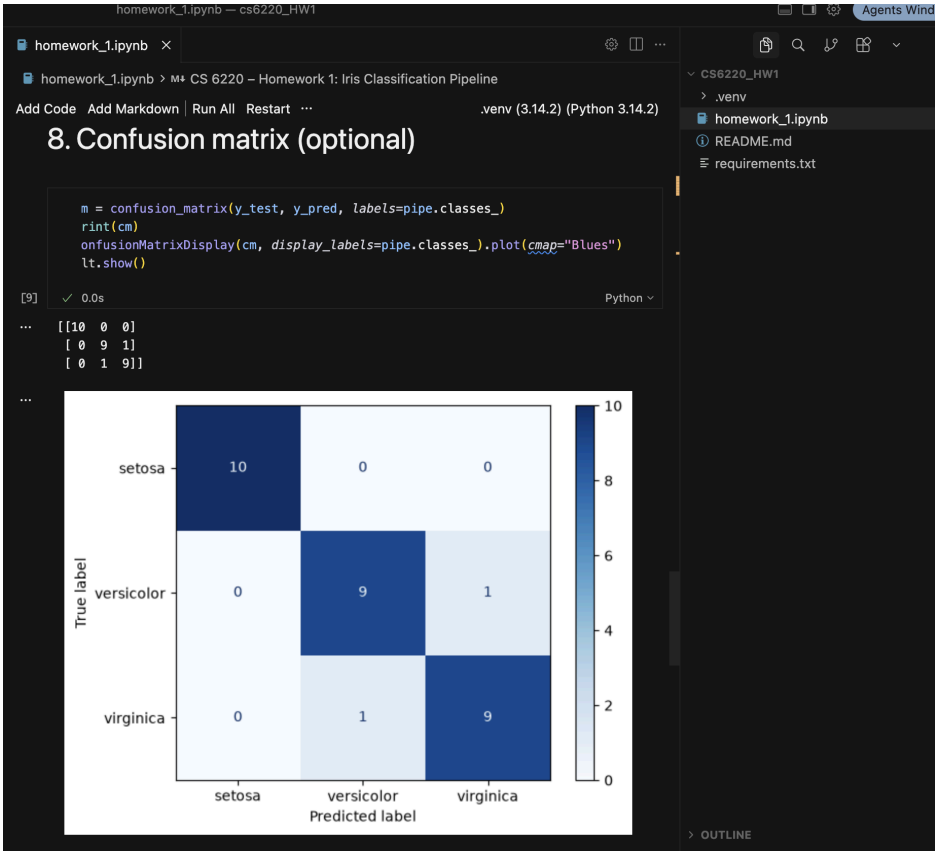


5) Program Execution

a)




b)




6) Results

a) Training time: 9.490 ms

b) Testing time: 0.935 ms

c) Test accuracy: 0.9333 (93.33%)

AI Usage

Prompts:

- I attached the homework PDF and said “This is my homework 1. Can you walk me through it, helping me with each step along the way.”
- I asked about screenshots.
- “What sort of questions could I be asking related to these results” which produced these questions which I then went through Claude with.
    - "Explain why the StandardScaler is inside the Pipeline instead of being applied before the train/test split. What goes wrong otherwise?"
    - "In my results, accuracy went from 77% at C=0.001 to 100% at C=10. Explain what C does in logistic regression and why accuracy changed."
    - "My test set has only 30 samples. How reliable is a 93% accuracy, and how would cross-validation give a better estimate?"

I used Claude to generate code, then I ran it, set up the github repo and made this report.